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Assignment 1

Elementary Programming in CiviC

This assignment series is supposed to familiarise yourself with the CiviC model programming language and to provide you with an initial test suite of example programs for your own CiviC compiler.

Assignment 1.1: CiviC Core Programming

Implement a CiviC module `core.cvc` that exports the following functions:

- `gcd(a,b)`
  returns the greatest common denominator of *a* and *b*;
- `fac(n)`
  returns the factorial of *n*;
- `fib(n)`
  returns the *n*-th Fibonacci number;
- `isprime(n)`
  returns `true` if *n* is a prime number and `false` otherwise.

Assignment 1.2: CiviC Nested Functions and I/O

Implement a CiviC module `coreio.cvc` that exports the following functions:

- `fibs(n)`
  print first *n* Fibonacci numbers;
- `primes(n)`
  print first *n* prime numbers.

The function `fibs` must make use of the function `fib` from the `core.cvc` module. In contrast, the function `primes` shall have a clone of function `isprime` as a nested local function definition.

Assignment 1.3: CiviC Arrays

Implement a CiviC module `array.cvc` that exports the following functions:

- `printIntVec( int[`*n*`] `*vec*`)`
  `printFloatVec( float[`*n*`] `*vec*`)`
  print *vec* to stdout;
- `printIntMat( int[`*m*`,`*n*`] `*mat*`)`
  `printFloatMat( float[`*m*`,`*n*`] `*mat*`)`
  print *mat* to stdout;
- `scanIntVec( int[`*n*`] `*vec*`)`
  `scanFloatVec( float[`*n*`] `*vec*`)`
  scan *vec* from stdin;
- `scanIntMat( int[`*m*`,`*n*`] `*mat*`)`
  `scanFloatMat( float[`*m*`,`*n*`] `*mat*`)`
  scan *mat* from stdin;
- `matMul( float[`*m*`,`*n*`] `*a*`, float[`*o*`,`*p*`] `*b*`, float[`*q*`,`*l*`] `*c*`)`
  multiply two floating point matrices *a* and *b* and store result in *c*;
- `queens( bool[`*m*`,`*n*`] `*a*`)`
  solve the well known 8-Queens problem (bonus challenge).

Note:

In the absence of characters and character strings in CiviC, your formatting options are very limited. Make the best out of it.

Note:

All above CiviC modules **must not** export a `main` function. For testing purposes write separate modules that do contain `main` functions with suitable but minimal test code and submit them alongside.

Assignment due date: Friday, November 6, 2026